

Transformers and Text Processing

IA Collaboration

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Text Embeddings (Token Embedding)

- ▶ Allows us to manipulate the representations of words in meaningful ways[cite: 60].
- ▶ We can find a word that is like another, or blend two words to find one that's in between them[cite: 61].
- ▶ This concept is key to developing attention, and then transformers[cite: 62].

Embeddings Arithmetic

Many semantic and syntactic relationships between words are (almost) linear in the word vector space[cite: 89]:

- ▶ **Semantic:** $v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$ [cite: 90, 98].
- ▶ **Syntactic:** $v(\text{kings}) - v(\text{king}) + v(\text{queen}) \approx v(\text{queens})$ [cite: 91].

⁰Mikolov, Tomas, et al. "Efficient estimation of word representations in vector space." arXiv:1301.3781 (2013)[cite: 100].

Embedding Words

- ▶ A word embedding represents a word as a vector of hundreds of dimensions[cite: 106].
- ▶ Instead of a single number, the algorithm assigns a list of numbers representing coordinates in a huge space[cite: 107].
- ▶ The system processes one-dimensional tensors (lists) instead of zero-dimensional tensors[cite: 114].

⁰El Boukkouri, Hicham (2018). "Arithmetic Properties of Word Embeddings" [cite: 109].

⁰spaCy authors (2020). "Word Vectors and Semantic Similarity" [cite: 110].

Comparing Embeddings by Similarity

Once we have embeddings, it is easy to incorporate them into almost any network and compare similarity[cite: 112, 139].

- ▶ Common algorithms: GLoVe (2014), Word2vec (2013), fastText (2020)[cite: 142, 144, 148].
- ▶ Universal Sentence Encoder (2018) for full sentences[cite: 188].

⁰Cer, Daniel, et al. "Universal Sentence Encoder." arXiv:1803.11175 (2018)[cite: 188].

ELMo (Embeddings from Language Models)

Problem: Many languages have words with different meanings (homonyms) written the same way[cite: 199].

- ▶ Example: "train" as a noun ("rode a train") vs. "train" as a verb ("train dogs")[cite: 203].
- ▶ **ELMo (2018):** The first algorithm to produce contextualized word embeddings[cite: 206, 207].

⁰Peters, Matthew E., et al. "Deep Contextualized Word Representations." arXiv:1802.05365 (2018)[cite: 226].

ELMo Visualization

ELMo places each word into a space of 1,024 dimensions based on context[cite: 237].

- ▶ We usually place embedding algorithms like ELMo as the very first layer in a language processing network[cite: 238, 239].
- ▶ It successfully distinguishes between verbal uses ("train a dog") and noun uses ("the train was empty")[cite: 235, 236].